Robotics & Automation · Multi-sensor System

Multi-sensor Registration for an Autonomous Forklift

Apply the coordinate registration used in surgical navigation between the robot, cameras, and sensors, and connect perception to safety decisions and the vehicle system.

Period 2025.11 - present | Evidence state Ongoing · Customer-site trial operation | Portfolio tier Robotics & Automation

Connected a Jetson-based cart recognition, classification, and action workflow (SAM3 segmentation, three-stage ToF-RGB calibration, DINOv2 cart classifier, vision servoing), sensor integration, and the initial safety-manager design into one flow on Zenoh-based in-house middleware. It moved from a Windows prototype (2026.02) to the team repository (2026.04), the Set3 baseline (2026.07), and the Jetson operations repository, and is now in customer-site trial operation (customer not named).

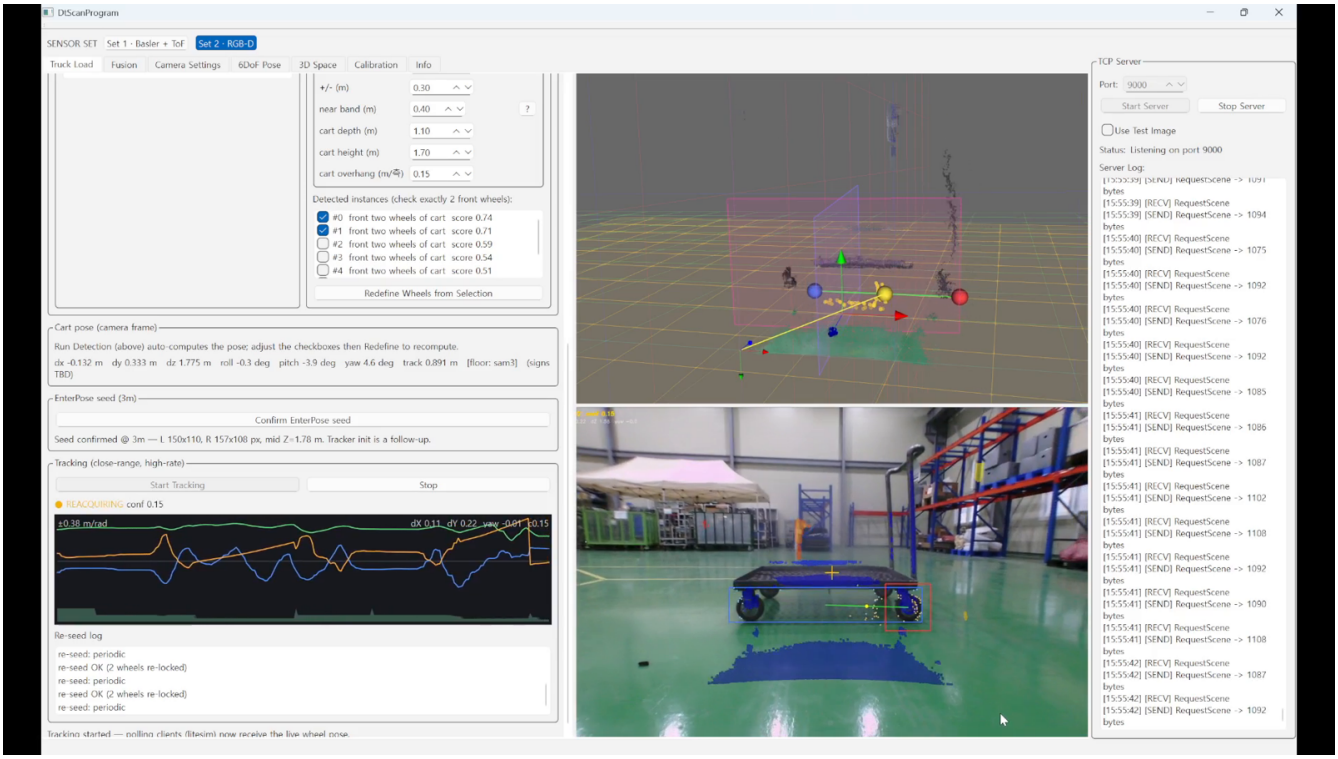


Figure 1. Screen recording of the DOTORI vision-servoing test (3D view, camera feed, signal plots).

Problem and system boundary

Sensor streams arrived in different coordinates and cycles and needed a common flow usable by safety decisions.

Owned decisions and implementation

My role

Owned the AI-vision cart recognition, classification, and action workflow. For recognition, the Jetson vision service (dotori_vision) uses SAM3 "cart/wheel/floor" prompt segmentation to get the wheel-pair midpoint pose, tied together by a three-stage ToF-RGB calibration chain (RGB distortion, ToF↔RGB, camera↔fork) on a 25×25 ChArUco board; the fork-board tilt check went from 5.51° to 0.69°. The cart classifier crops the cart with the same mask, extracts DINOv2-S features, and returns the nearest-gallery top-3, mapping 24 cart types onto 8 control profiles (offline top-1 99.5% on 2,010 queries from phone video of 24 types; robot-camera queries are not yet validated; Basler-style augmentation is used, and the DINOv3 comparison is research only). The operator confirms among three PDA candidate images, and the chosen profile values drive the measure, align, insert, and lift actions. The floor plane is handled by an A/B comparison of the SAM3 mask against ToF RANSAC, a floor-reflection signed-height gate, and fail-close on inner-caster mis-selection; a RequestImage/RequestPointCloud TCP API is provided. Implemented the vision-servoing actions (approach_cart_rgbd, check_fork_alignable) and integrated them into the team's behaviour trees; approach_plane_lidar/tof is at the development and experiment stage (one real-vehicle run against a wall). The Isaac Sim digital twin (RTX LiDAR, CAN simulation, operator console, full RCS stack end-to-end drive_arc, July-August 2026) is an experimental prototype. Built the sensor-integration service with a fused 3D occupancy output and mast-LiDAR surface detection, and designed and initially implemented the safety manager (per-sensor warning and stop zones with E-STOP latching), which the team later extended into per-action safety profiles. Map and mission tooling is integrated with the field tools the team built.

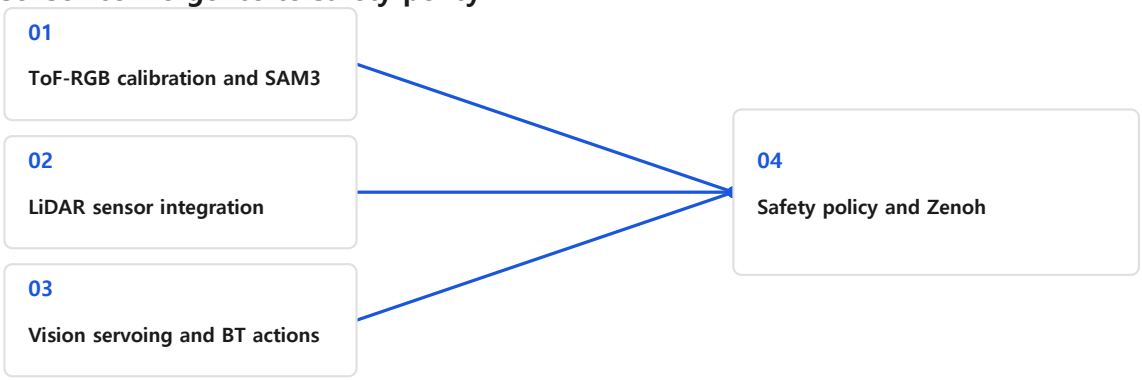
Sensor coordinate chain

Connected a three-stage calibration chain (RGB distortion, ToF↔RGB, camera↔fork; 25×25 ChArUco board, fork-board tilt 5.51° to 0.69°) and the mast LiDAR to vehicle coordinates, and produced one 3D occupancy output through the sensor-integration service. The sensors are three safety 2D LiDARs, a mast LiDAR, a reflector-based positioning scanner, a ToF camera (Basler blaze-102), and RGB.

Perception to policy

Published the results of the chain from SAM3 recognition to DINOv2 classification (top-3), operator confirmation, and profile-based actions over Zenoh, integrated vision-servoing and loading actions into the team's behaviour trees, and drew up the safety manager's initial design as per-sensor warning and stop zones with E-STOP latching, which the team later extended into per-action safety profiles.

Multi-sensor convergence to safety policy



Explanatory system relationship diagram; it is not photographic or experimental evidence.

Verification and evidence ledger

Test-ground recordings and photographs, capture-specific sensor comparisons, and my own field-trial measurements provide the evidence. The integration-evidence section below states the timestamp and extrinsic-measurement conditions and differences by axis.

Integration evidence

In a 191-frame, 9.7399-second capture on 29 July 2026, the ToF↔RGB timestamp difference had a p95 of 16.592 ms (16.6 ms rounded). This is not end-to-end system latency. A single-frame extrinsic-translation comparison on 11 September was within 1 mm of the mechanical model in x and y, but differed by about 29 mm in z; it does not establish overall 3D accuracy. My own measurements: offline classification top-1 99.5% on 2,010 queries (phone video, 24 types); in the field trial, a front 2 m measurement of 4.66 s (median of 10) and a track read of 0.71 m against 0.75 m by tape. The measurable range of about 1.5-5 m (limit) is the operating limit of the vision measurement, not a measured result.

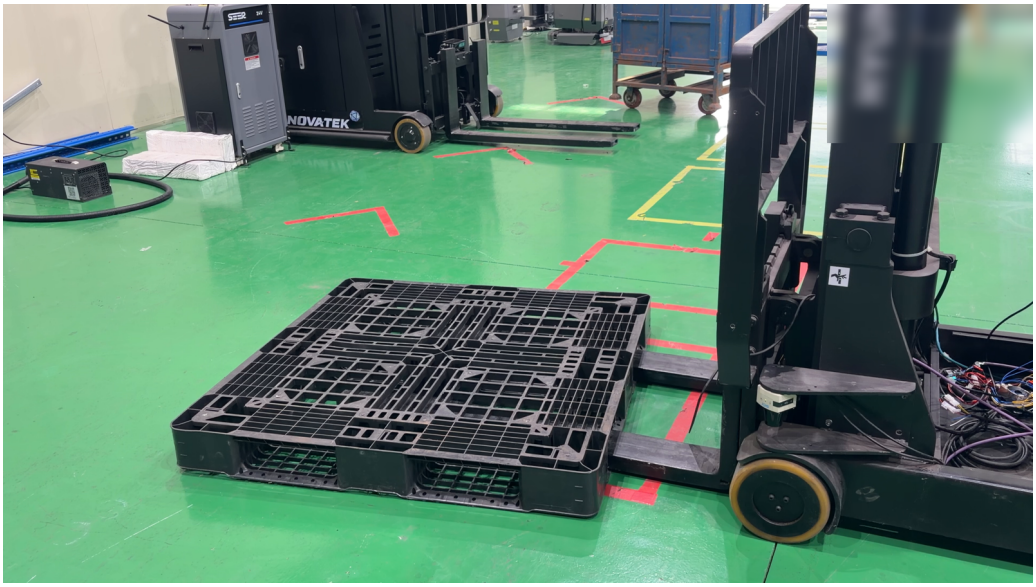


Figure 2. Fork alignment just before pallet entry during the full-scenario test



Figure 3. Trajectory-following drive test, wide view of the test site



Figure 4. Fork-lift test lifting a cart

Role, team result, and limits

Team result

Localisation (including the reflector-based positioning scanner), vehicle control and hardware abstraction, mission management, the behaviour-tree framework, the point-cloud publisher, and later safety-manager policy extensions were owned by team members; the team performed system integration and field validation. This is not presented as production operation, deployment success, or customer outcomes.

Limitations

The public clip and photos show tests at the test ground. The 4.66 s and 0.71 m values are my own field-trial measurements, and the classification accuracy is based on offline phone video. No production or customer outcomes are claimed.

Field boundary

In earlier forklift field work during 2025, I supported operation, issue observation, and handovers. The public record of participation in the in-house DOTORI project begins in November 2025, followed by implementation records in the current repositories during 2026. Field support and later vision/sensor implementation are distinct; test-site validation is not a production or customer outcome.

Collaboration boundary

Vehicle-control, localisation, mission, safety, and field-validation owners integrate the system jointly. The middleware is an in-house Zenoh-based implementation that follows ROS 2-style patterns.

Implementation technologies

C++ · Zenoh · BehaviorTree.CPP · ToF · RGB · SAM3 · Jetson AGX Orin · 2D LiDAR · DINOv2 · Isaac Sim

Primary capability

Robot Vision & Sensor Integration · 3D Registration & Computer Vision

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Registered public evidence

No external public link is currently registered.

What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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